era when much of the tech-driven world is heavily dependent on digitisation, cyber security is an absolute necessity.

WHY IT IS A GAME CHANGER

With most defence information/ plans/ electronic systems digitised, a breach of cyber security can endanger the nation's closely-guarded secrets and compromise its security. India's adversaries, and China in particular, will be trying to spy on our defence installations through Chinese-made CCTV cameras and through outright efforts to hack important ministries and organisations like the Defence Research and Development Organisation. Terrorist organisations, too, are a constant threat to cyber security.

To counter such cyber warfare, India has a multi-tiered cyber security system. Set up in 2004, the Indian Computer Emergency Response Team (CERT-IN) is the nodal agency to deal with cyber security threats. The National Cyber Security Protection Act, 2014 established national cyber security centres and methods to protect critical infrastructure.

The National Critical Information Infrastructure Protection Centre (NCIIPC) also protects assets of critical sectors like energy, banking, telecom and defence. In 2018, the National Cyber Coordination Centre (NCCC), a cyber security and e-surveillance agency of the government, was established. Other than this, the private sector too has joined in. Several companies have set up their own security operations centres (SOCs) to monitor and respond to cyber attacks. As the threat of cyber crimes and warfare is constant, all these can be transformative if they work well.

CHALLENGES

- No centralised, national-level cyber security architecture
- Lack of awareness as a national cyber security regulatory policy does not exist for individuals
- Insufficient resources to address cyber security concerns

THE CHANGE-MAKERS

CLOUDSEK
Bengaluru-based AI company that predicts cyber threats

ARCON TECH SOLUTION
A Mumbai-based firm, it specialises in the enterprise identity security domain

BLUESAPPHIRE CYBER SYSTEMS
The Hyderabad-based firm addresses cybersecurity challenges for enterprises across industries

DATA RESOLVE TECHNOLOGIES
The Noida-based company with expertise in internal data management and cyber security

1,725
Number of cyber attacks faced every week by India's defence establishment and other government departments in January-March, 2023



"The current state of Indian cyber security is like...many small kingdoms with their own defence strategies. The landscape requires a unifying policy to secure them"

– RAHUL SASI
CEO & Co-founder, CloudSEK

WHAT INDIA NEEDS TO MASTER

Organisations typically targeted by hackers and cyber criminals in India reported 1,787 cyber attacks a week on average in the last six months, as compared to the global average of 983, according to cyber security solutions company Check Point Software Technologies. There has to be more awareness about cyber crimes, for a majority of India's population is digitally literate but unaware of basic security measures.

The financial sector, too, needs to safeguard information and be vigilant of ATM systems being hacked to account-centric frauds. Moreover, the military's cyber security is crucial. The armed forces must develop capabilities to send secure messages and make secure calls. A process to send real-time alerts to key users of systems/ networks must be developed. Reporting of real-time incidents is also of great importance. ■